

Hypotenuse



1 For each of the following, find the number that completes a Pythagorean triple given the two smallest numbers:

a 3 and 4

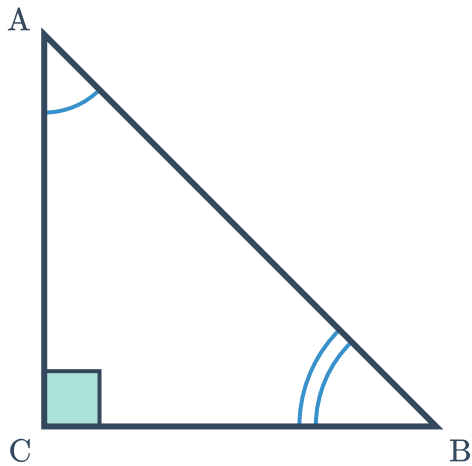
b 8 and 6

c 9 and 12

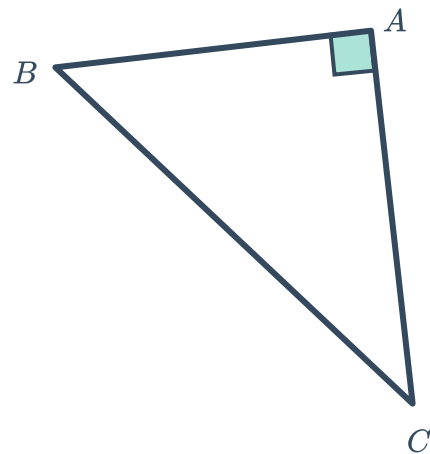
d 8 and 15

2 For each diagram, identify which side of the triangle is the hypotenuse.

a



b

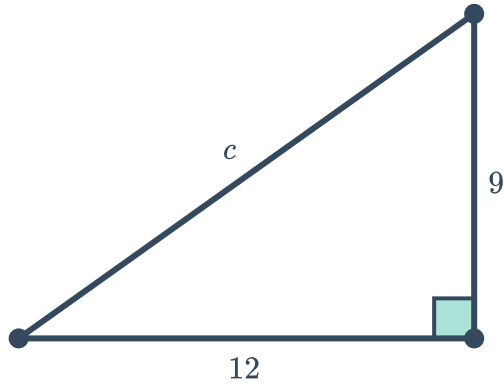


3 For the following right-angled triangles:

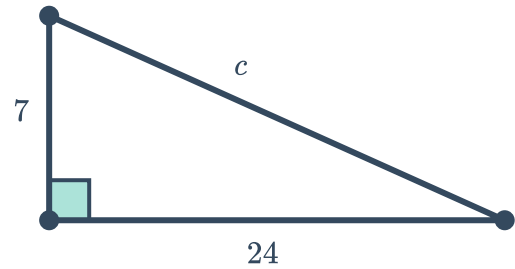


- i Write the equation that the sides of the triangle satisfy.
- ii Solve the equation to find the length of the hypotenuse.

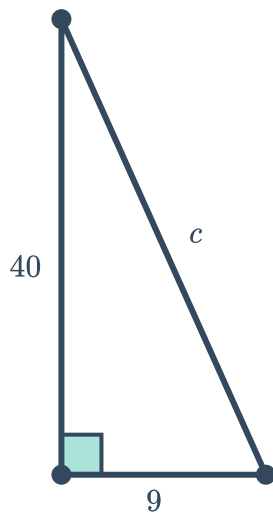
a



b



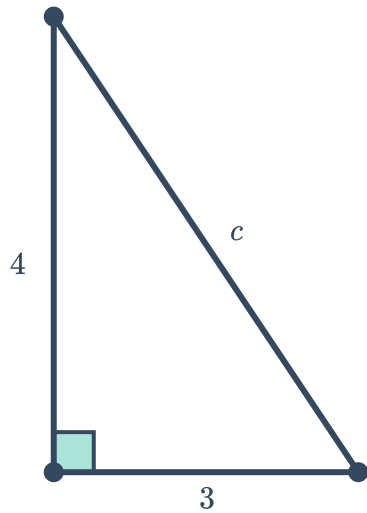
c



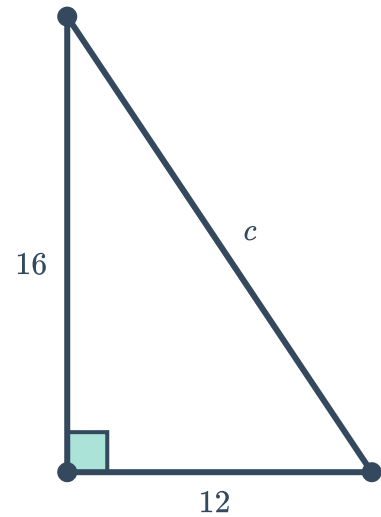
4 Find the length of the unknown side of the following triangles:



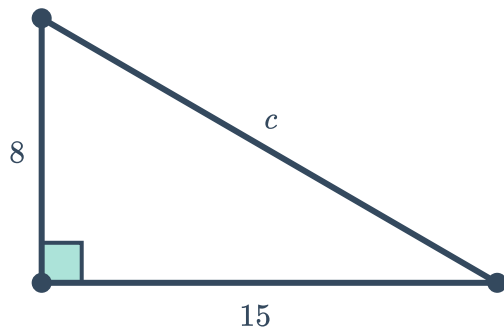
a



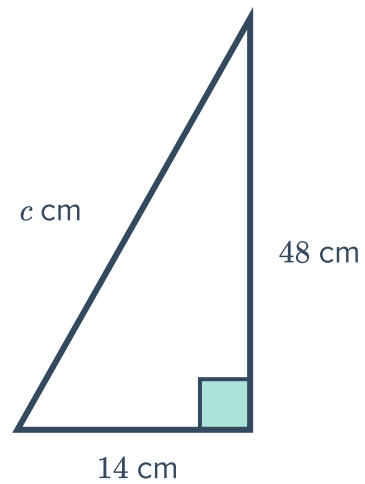
b



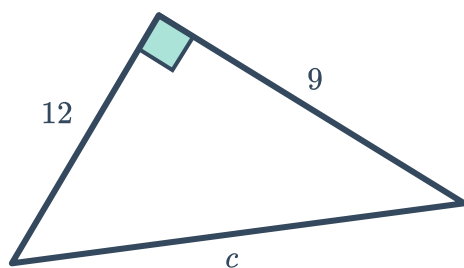
c



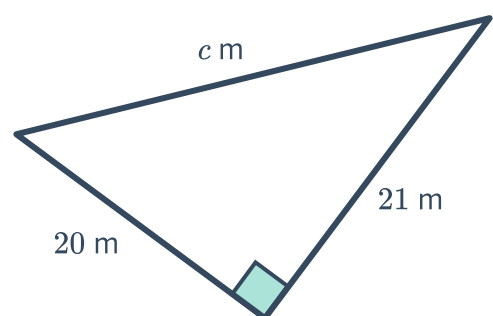
d



e

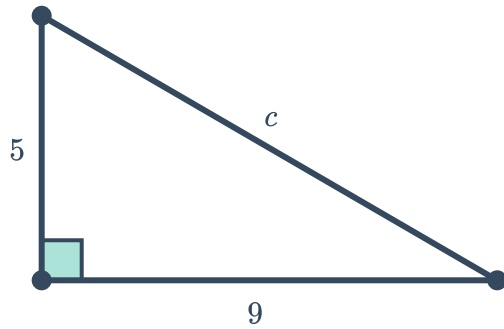


f

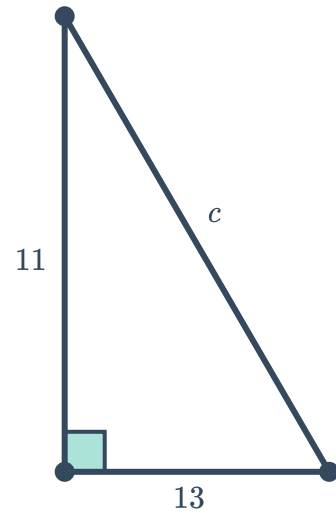


5 Find the length of the unknown side of the following triangles as a surd:

a

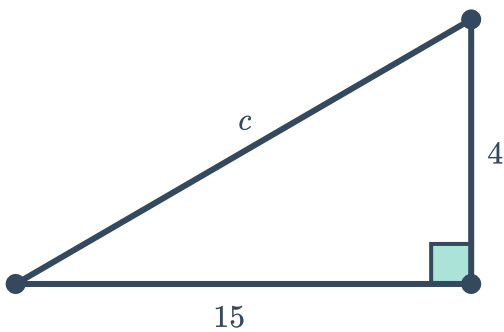


b

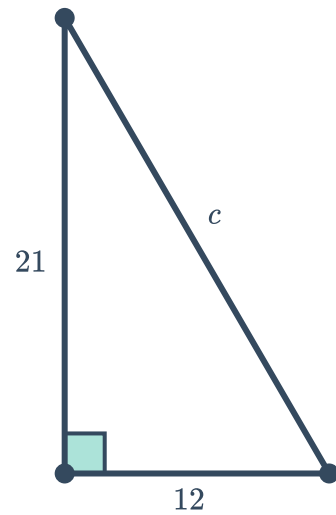


6 Find the length of the unknown side of the following triangles, correct to two decimal places:

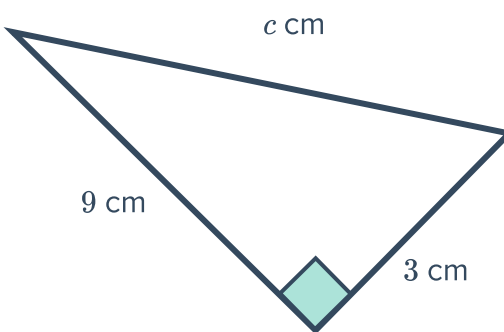
a



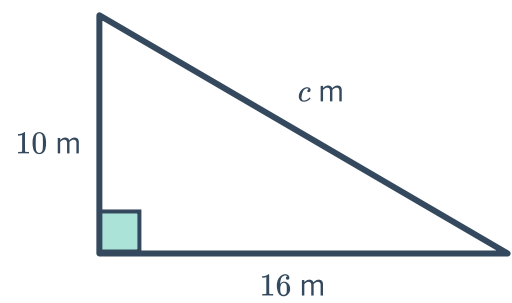
b



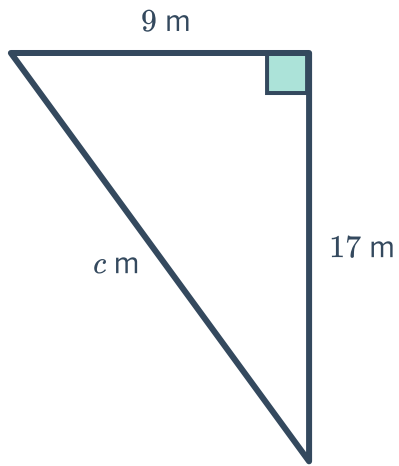
c



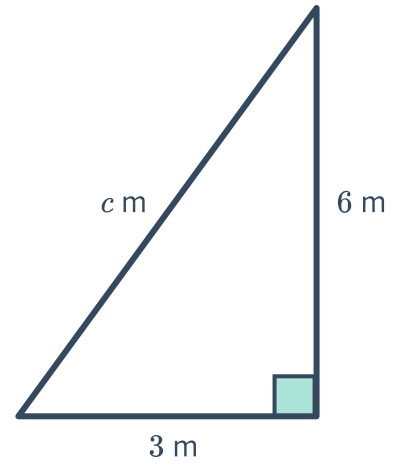
d



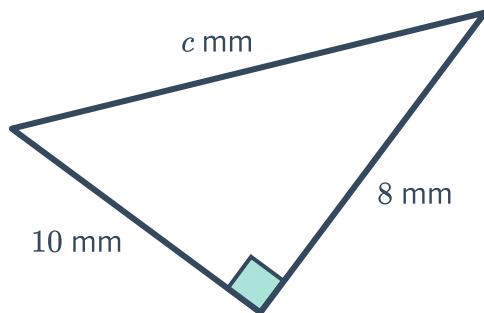
e



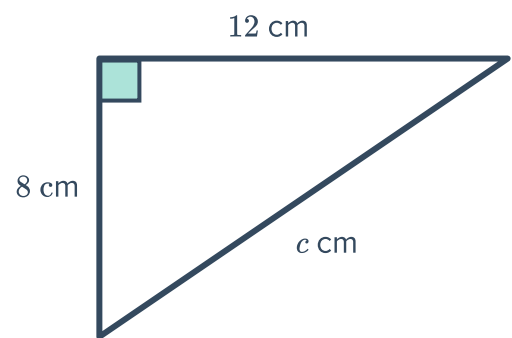
f



g



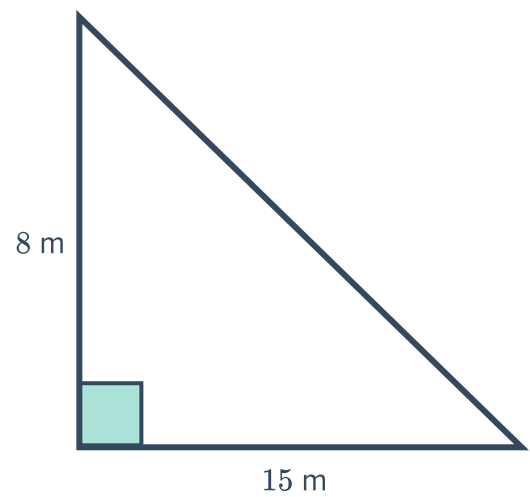
h



Perimeter

7 Consider the following triangle:

- a Find the length of the hypotenuse.
- b Find the perimeter of the triangle.





- a Find the length of the hypotenuse, correct to two decimal places.
- b Find the perimeter of the triangle, correct to two decimal places.

- 9 A right isosceles triangle has two sides measuring 17 cm.
- a Find the length of its hypotenuse, correct to two decimal places.
 - b Find the perimeter of the triangle, correct to two decimal places.

Diagonals of rectangles

- 10 Find the exact length of the diagonals of the following rectangles:
- a A rectangle whose side lengths are 9 and 12.
 - b A rectangle whose side lengths are 9 and 14.
- 11 Find the length of the diagonals of the following squares, correct to two decimal places:
- a A square with side length of 9.
 - b A square with side length of 12.

- 12 Complete the following table:

Side length of square	Exact length of diagonal
1	
3	
6	
11	
249	